

Several Simple and Precise Quantitative Equations for the 3D Coordinates of Main-Chain Backbone Atoms in Protein Structure

Prediction

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Abstract: By studying experimentally determined protein structure data downloaded from the world-renowned protein structure database, the Protein Data Bank (PDB), the author surprisingly discovered that the three-dimensional coordinates of atoms on the protein main-chain backbone obey several simple and precise physical equations. Assuming that one-direction pull forces between adjacent atoms on the protein main-chain backbone are of equal magnitude, the sums of the projections of these forces along the x -, y - and z -axes are all zero, while the sums of the corresponding torque projections are not zero. Based on these calculations, we propose that main-chain backbones of proteins belong to "force balance and torque imbalance" systems in the context of Engineering Mechanics.

Keywords: Protein structure prediction; force balance; torque imbalance; three-dimensional coordinates; quantitative equation

Proteins are the fundamental building blocks of life, participating in nearly all biological processes—from enzymes that catalyze chemical reactions and receptors that transmit signals, to structural proteins that provide cellular support. The diversity and functionality of proteins make them core molecules for sustaining life activities. Proteins are folded from chains of amino acids, and their three-dimensional structures determine their functions. Therefore, understanding protein structure is of critical importance for biological research, disease treatment, and even industrial applications¹. Traditional experimental methods include X-ray crystallography, nuclear magnetic resonance (NMR) spectroscopy, and cryo-electron microscopy (cryo-EM). However, these methods are time-consuming and expensive²⁻³, which limits their practicality for large-scale applications and means that the number of experimentally determined protein structures still falls far short of the demands of modern science. Consequently, researchers have long sought to predict protein three-dimensional structures through computational approaches. Nevertheless, traditional protein structure prediction algorithms, which are primarily based on homology modeling of known structures and prior physical knowledge,

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suffer from low accuracy and produce results that deviate significantly from experimentally determined data³. Moreover, even the currently most prominent model, AlphaFold from Google, which employs AI technology and demonstrates significantly higher accuracy than most other prediction methods⁴, has been argued by some biologists as a "black-box" model with insufficient physical interpretability and mechanistic clarity⁵⁻⁶. Like other existing prediction methods, it does not truly address the underlying theory or mechanism of protein folding⁷⁻⁹. While studying experimentally determined data downloaded from the world-renowned protein structure database, Protein Data Bank (PDB), the author unexpectedly discovered that, assuming equal-magnitude one-direction pull forces between adjacent atoms, there exist simple and precise quantitative relationships among the three-dimensional coordinates of different atoms on the protein's main chain backbone. This paper briefly introduces our work by using single-chain proteins as examples, with the aim of providing new insights and methods for protein structure prediction work.

1. Theoretical Section

1.1 Force Decomposition

If a force f_i is directed from point A to point B, with the three-dimensional coordinates of points A and B being (x_i, y_i, z_i) and $(x_{i+1}, y_{i+1}, z_{i+1})$, respectively, then the projection of this force along the x -axis, F_{xi} is given by¹⁰ (for the sake of clarity, only the x -axis is discussed below; the cases for the y - and z -axes can be derived analogously):

$$F_{xi} = \frac{x_{i+1} - x_i}{\sqrt{(x_{i+1} - x_i)^2 + (y_{i+1} - y_i)^2 + (z_{i+1} - z_i)^2}} f_i \quad (1)$$

For a protein main-chain backbone containing $n+1$ atoms to be studied from left to right, we assume that there exist one-direction pull forces between any two adjacent atoms on the main-chain backbone (all pull forces towards to one end of protein main-chain backbone like on a taut light string), then the x -axis projections of all pull forces can be expressed using Equation (1). The sum of their x -axis projections is then:

$$F_x = \sum_{i=1}^n \frac{x_{i+1} - x_i}{\sqrt{(x_{i+1} - x_i)^2 + (y_{i+1} - y_i)^2 + (z_{i+1} - z_i)^2}} f_i \quad (2)$$

1.2 Force Balance and Torque Balance Conditions

1.2.1 Force Balance Condition

According to Engineering Mechanics, if the sums of the projections of a force system along the x -, y -, and z -axes are all zero, i.e., $F_x = 0$, $F_y = 0$, and $F_z = 0$, the force system is in a state of force balance¹¹. Assuming all pull forces f_i between adjacent atoms are of equal magnitude, Equation (2) becomes:

$$F_x = f \sum_{i=1}^n \frac{x_{i+1} - x_i}{\sqrt{(x_{i+1} - x_i)^2 + (y_{i+1} - y_i)^2 + (z_{i+1} - z_i)^2}} \quad (3)$$

Since the magnitude of the pull force f is not zero, the condition for force balance along the x -axis can be transformed into:

$$\Delta_x = \sum_{i=1}^n \frac{x_{i+1} - x_i}{\sqrt{(x_{i+1} - x_i)^2 + (y_{i+1} - y_i)^2 + (z_{i+1} - z_i)^2}} = 0 \quad (4)$$

Considering experimental measurement errors, the use of a ratio expression can better reflect the actual situation:

$$r_x = \sum_{i=1}^n \frac{x_{i+1}}{\sqrt{(x_{i+1} - x_i)^2 + (y_{i+1} - y_i)^2 + (z_{i+1} - z_i)^2}} / \sum_{i=1}^n \frac{x_i}{\sqrt{(x_{i+1} - x_i)^2 + (y_{i+1} - y_i)^2 + (z_{i+1} - z_i)^2}} = 1 \quad (5)$$

1.2.2 Torque Balance Condition

Torque is a physical quantity that describes the rotational effect of a force on an object, and its magnitude is equal to the product of the force and the moment arm. Similarly, for a force directed from point A to point B, the torque about an arbitrary point in space is calculated as follows¹¹:

$$\begin{aligned} M_{xi} &= y_i' F_{zi} - z_i' F_{yi} \\ M_{yi} &= z_i' F_{xi} - x_i' F_{zi} \\ M_{zi} &= x_i' F_{yi} - y_i' F_{xi} \end{aligned} \quad (6)$$

In Equation (6), x'_i , y'_i , and z'_i are the relative three-dimensional coordinates of the starting point A of the force with respect to an arbitrary point in space; F_{xi} , F_{yi} and F_{zi} are the projections of the force along the x -, y -, and z -axes, respectively; and M_{xi} , M_{yi} , and M_{zi} are the projections of the torque about the an arbitrary point along the x -, y -, and z -axes, respectively. Again taking the x -axis as an example, the sum of all M_{xi} of protein main-chain backbone is:

$$M_x = \sum_{i=1}^n (y'_i F_{zi} - z'_i F_{yi}) \quad (7)$$

Substituting the expressions for F_{yi} and F_{zi} , derived by analogy from Equation (1), into Equation (7). Again assuming that all one-direction pull forces between adjacent atoms are of equal magnitude, the condition for torque balance along the x -axis is:

$$M_x = f \sum_{i=1}^n \frac{y'_i (z_{i+1} - z_i) - z'_i (y_{i+1} - y_i)}{\sqrt{(x_{i+1} - x_i)^2 + (y_{i+1} - y_i)^2 + (z_{i+1} - z_i)^2}} = 0 \quad (8)$$

Since the magnitude of the pull force f between adjacent atoms is not zero, Equation (8) transforms into:

$$\Delta'_x = \sum_{i=1}^n \frac{y'_i (z_{i+1} - z_i) - z'_i (y_{i+1} - y_i)}{\sqrt{(x_{i+1} - x_i)^2 + (y_{i+1} - y_i)^2 + (z_{i+1} - z_i)^2}} = 0 \quad (9)$$

and the ratio expression:

$$R_x = \sum_{i=1}^n \frac{y'_i (z_{i+1} - z_i)}{\sqrt{(x_{i+1} - x_i)^2 + (y_{i+1} - y_i)^2 + (z_{i+1} - z_i)^2}} / \sum_{i=1}^n \frac{z'_i (y_{i+1} - y_i)}{\sqrt{(x_{i+1} - x_i)^2 + (y_{i+1} - y_i)^2 + (z_{i+1} - z_i)^2}} = 1 \quad (10)$$

The conditions for torque balance along the y - and z -axes can be derived analogously.

2. Calculation Results and Discussions

The experimentally determined three-dimensional coordinate data used in the calculations of this paper were selected from the world-renowned protein structure database, Protein Data Bank (PDB), accessible at www.rcsb.org. The x -, y -, and z -axis coordinates of the amide nitrogen atoms (N), alpha-carbon atoms ($C\alpha$), and carbonyl carbon atoms (C) of each amino acid residue on the main chain backbone were extracted from the downloaded PDB files in cmf format and substituted into the above formulas. The results are shown in Table 1.

Table 1. Calculation results of force balance and torque balance equations for some single-chain proteins

No	Species	Protein Name	PDB ID	Force Balance of Main-Chain backbone						Torque Balance of Main-Chain backbone					
				Atoms						Atoms					
				Δ_x	Δ_y	Δ_z	r_x	r_y	r_z	Δ'_x	Δ'_y	Δ'_z	R_x	R_y	R_z
1	Homo sapiens	C-Jun N-terminal kinase 3	2P33	-21.8181	-15.8099	-22.5539	0.998187	0.998531	0.998579	-61.463	846.668	-207.28	1.26488	0.106976	2.46604
2	Escherichia coli	GTP-binding protein	5A9W	-31.901	-24.2489	17.9143	1.00177	1.00044	1.00056	-372.429	-862.195	-1817.34	2.01795	3.67856	0.206295
3	Homo sapiens	Smad2 protein	1KHX	-8.51191	11.3153	-17.4289	0.999735	1.00029	0.999002	-1710.38	49.4962	1579.79	-2.93613	0.929162	-1.14267
4	Dictyostelium discoideum	STAT protein	1UUR	9.74338	-29.5661	-12.8342	1.00053	0.999275	0.998639	758.147	741.905	-1609.01	-0.62038	-0.91019	-1.63557
5	Elephas maximus	Myoglobin	1EMY	0.638443	-2.44847	-11.0165	1.00011	0.999742	0.998974	-727.968	159.974	-131.709	-4.52551	-0.93224	-0.84573
6	Homo sapiens	Enteropeptidase	7WQX	-4.22926	-5.9311	-33.8083	0.99997	0.999957	0.999751	-1035.8	3640.23	-535.741	1.4759	0.171124	2.32379
7	Homo sapiens	Hepatocyte growth factor	1SI5	7.27377	-0.11929	15.0224	1.00032	0.999345	1.00049	491.519	99.6795	604.046	-1.72591	1.15251	-0.79627
8	Homo sapiens	Glutathione reductase	1GRH	46.4453	-1.54993	-1.25798	1.00068	0.999969	0.99993	481.263	2508.42	-2622.22	-0.57704	3.42575	-0.11151
9	Homo sapiens	Myocilin	9DOZ	-2.61701	1.56349	4.7472	1.00012	1.00063	0.9995	162.199	1352.43	620.938	-0.05607	-0.88676	-1.14487
10	Equus caballus	Kallikrein-1E2	1GVZ	8.24975	17.4441	-7.90233	1.00121	1.00116	0.999259	-239.813	718.623	-376.496	-0.98741	-1.05305	0.206861
11	Aequorea victoria	Blue fluorescent protein	1BFP	8.54198	11.6162	-2.97218	1.00069	1.00086	0.999836	-529.868	86.5174	-325.787	0.023692	1.69595	0.243865
12	Homo sapiens	Steroid hormone receptor ERR1	3K6P	-10.1977	-6.75052	6.38339	0.999438	0.998705	1.00044	1234.07	-579.181	-250.44	-1.18589	-1.30924	3.05024
13	Homo sapiens	Prostasin	3DFJ	14.3317	-22.851	11.1827	0.999055	1.00272	1.00103	694.345	48.3546	580.684	-0.42773	-3.22234	-2.16734
14	Salmonella typhimurium	65 kDa heat shock protein	2GWL	9.42642	9.87651	-10.6013	1.00091	1.00104	0.998868	-362.502	193.361	-733.235	-0.34879	-0.46492	-0.21343

15	Anser anser anser	Lysozyme	154L	-5.36189	-5.52789	5.02461	0.997371	0.999531	0.999588	-540.029	-306.903	951.17	-8.81242	-8.00787	-0.65045
16	Mus musculus	Neuroglobin	1W92	-11.278	15.9262	4.11257	0.997955	1.00111	1.00041	-703.908	-667.504	1005.44	0.134145	-5.21195	-0.70857
17	Homo sapiens	Growth hormone	1HGU	12.6704	-1.61937	4.03496	1.00066	0.999887	1.0002	-217.586	810.179	-53.9577	-0.4023	10.6871	0.867056
18	Cannabis sativa	Tetrahydrocannabinolic acid synthase	3VTE	11.3546	7.67779	18.011	1.00016	1.00012	0.999378	651.423	-1716.72	-1154.84	2.92485	-0.51902	0.178115
19	Mus musculus	Cytosin 2	1U27	-2.22643	-10.3283	0.920846	0.999418	0.998188	1.00011	573.518	-351.358	135.867	-0.36191	-1.40277	3.31252
20	Homo sapiens	Plasminogen	1RJX	19.361	0.285419	-3.35949	1.00028	1 . 0000	0.999851	264.45	1295.8	-2765.8	0.40311	-1.8051	0.025646
21	Homo sapiens	Angiostatin	1KI0	-27.4532	-7.29566	3.61359	0.998052	0.999592	1.00031	1117.95	-1830.77	593.872	-0.61738	-9.95712	0.354456
22	Oncorhynchus keta	Trypsin	1MBQ	-21.8058	-1.06258	4.94438	0.996868	0.999277	1.00051	-413.169	-81.1307	-30.6884	-1.06772	1.4960	1.85275

Note: For computational convenience, the origin of the coordinate system was used as the center of moments in the torque calculations.

As can be seen from the calculation results in Table 1, the force balance conditions are satisfied very well for all the proteins examined, with the ratio values being nearly equal to 1 in all cases. In contrast, none of the torque balance conditions are satisfied. This indicates that main-chain backbones of proteins are "force balance and torque imbalance" systems in the context of Engineering Mechanics.

This paper is the first brief report in a series of studies by the author on protein structure prediction. The theoretical justification for assuming equal-magnitude one-direction pull forces between adjacent atoms on protein the main-chain backbone will be discussed in a separate paper.

Protein structure prediction is a fundamental problem in the life sciences. Capturing the underlying laws of protein folding and thereby achieving accurate prediction of protein structures has long been one of the most important challenges in structural biology¹². In this sense, the simple and precise formulas and regularities governing the three-dimensional coordinates of protein main-chain backbone atoms discovered in this work are undoubtedly beneficial.

Computer codes used in this study are available at: <https://github.com/tongji92774>

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